Table A4. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), multi-vehicle crashes without a motorcycle, Thailand, 2016–2023 (yearly periods), partially constrained temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016–2017]	–0.296	–2.44		
Constant [2018–2021]	–0.754	–7.39		
Constant [2022]	–0.898	–6.81		
Constant [2023]	–0.746	–5.85		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
One pickup truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.489	–8.37	0.0417	–0.0417
Two or more pickup trucks involved (1 if yes, 0 otherwise) [2016–2023]	–1.759	–11.04	0.0922	–0.0922
(Standard deviation of parameter distribution; 13.1% above zero)	1.568	8.43		
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.394	–3.00	–0.0010	0.0010
(Standard deviation of parameter distribution; 39.9% above zero)	1.546	6.29		
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.383	–2.57	–0.0090	0.0090
(Standard deviation of parameter distribution; 41.2% above zero)	1.725	6.10		
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	–0.255	–2.31	–0.0291	0.0291
(Standard deviation of parameter distribution; 43.6% above zero)	1.589	8.69		
Two or more trucks with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	–0.439	–3.59	0.0350	–0.0350
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2023]	–0.430	–7.95	0.0372	–0.0372
Crash cause				
Driver fell asleep (1 if yes, 0 otherwise) [2016–2023]	0.688	6.79	–0.0664	0.0664
Roadway characteristics				
T-intersection (1 if yes, 0 otherwise) [2016–2023]	–0.205	–2.12	0.0171	–0.0171
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise) [2016–2023]	0.255	1.66	–0.0229	0.0229
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2016–2023]	–0.445	–5.97	0.0359	–0.0359
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2023]	0.412	7.39	–0.0364	0.0364
Temporal characteristics				
Weekend (1 if yes, 0 otherwise) [2016–2023]	0.079	1.68	–0.0068	0.0068
Midnight / early morning (1 if yes, 0 otherwise) [2016–2023]	0.446	7.02	–0.0406	0.0406
Variables with temporally varying parameters				
Vehicle involvement				
One passenger car involved (1 if yes, 0 otherwise) [2016–2017]	–0.882	–8.64	0.0884	–0.0884
One passenger car involved (1 if yes, 0 otherwise) [2018–2023]	–0.710	–10.75	0.0569	–0.0569
Two or more passenger cars involved (1 if yes, 0 otherwise) [2016–2017]	–1.759	–12.84	0.1600	–0.1600
Two or more passenger cars involved (1 if yes, 0 otherwise) [2018–2019]	–1.129	–9.12	0.0986	–0.0986
Two or more passenger cars involved (1 if yes, 0 otherwise) [2020–2023]	–1.438	–12.96	0.0929	–0.0929
Bus involved (1 if yes, 0 otherwise) [2021–2022]	0.556	1.70	–0.0462	0.0462
Bus involved (1 if yes, 0 otherwise) [2023]	–0.796	–1.94	0.0480	–0.0480
Crash type				
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2016–2019]	–0.821	–7.33	0.0726	–0.0726
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2021–2023]	–2.671	–3.86	0.0423	–0.0423
(Standard deviation of parameter distribution; 22.0% above zero)	3.454	4.84		
Head-on collision (1 if yes, 0 otherwise) [2016–2020]	1.325	13.54	–0.1554	0.1554
Head-on collision (1 if yes, 0 otherwise) [2021–2023]	0.969	8.89	–0.0829	0.0829
Crash cause				
Speeding (1 if yes, 0 otherwise) [2021–2023]	0.292	3.78	–0.0209	0.0209
Cutting in too closely (1 if yes, 0 otherwise) [2016–2021]	–0.813	–3.64	0.0173	–0.0173
(Standard deviation of parameter distribution; 33.5% above zero)	1.904	5.58		

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Cutting in too closely (1 if yes, 0 otherwise) [2022]	0.422	2.38	–0.0330	0.0330
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2017]	–0.570	–3.73	0.0543	–0.0543
Road under construction (1 if yes, 0 otherwise) [2016–2018]	0.508	2.34	–0.0573	0.0573
Road under construction (1 if yes, 0 otherwise) [2019]	1.450	3.95	–0.1626	0.1626
Road under construction (1 if yes, 0 otherwise) [2021–2022]	0.551	2.46	–0.0456	0.0456
Painted median (1 if yes, 0 otherwise) [2016–2020]	0.390	3.97	–0.0395	0.0395
Raised or earth median (1 if yes, 0 otherwise) [2016–2018]	–0.487	–5.28	0.0483	–0.0483
Raised or earth median (1 if yes, 0 otherwise) [2021–2022]	–0.400	–3.91	0.0293	–0.0293
Raised or earth median (1 if yes, 0 otherwise) [2023]	–0.681	–4.67	0.0451	–0.0451
Depressed median (1 if yes, 0 otherwise) [2016–2018]	–0.425	–4.99	0.0426	–0.0426
Depressed median (1 if yes, 0 otherwise) [2019–2023]	–0.595	–4.17	0.0205	–0.0205
(Standard deviation of parameter distribution; 32.7% above zero)	1.329	4.92		
Barrier median (1 if yes, 0 otherwise) [2016–2017]	–1.397	–9.46	0.1241	–0.1241
Barrier median (1 if yes, 0 otherwise) [2018]	–0.719	–4.40	0.0697	–0.0697
Barrier median (1 if yes, 0 otherwise) [2019–2020]	–1.713	–13.85	0.1309	–0.1309
Barrier median (1 if yes, 0 otherwise) [2021–2023]	–2.169	–18.06	0.1469	–0.1469
Horizontal curve (1 if yes, 0 otherwise) [2016]	0.526	2.96	–0.0590	0.0590
Horizontal curve (1 if yes, 0 otherwise) [2017–2023]	–0.537	–2.87	–0.0149	0.0149
(Standard deviation of parameter distribution; 40.6% above zero)	2.257	6.32		
Concrete pavement surface (1 if yes, 0 otherwise) [2017–2023]	–0.212	–3.12	0.0173	–0.0173
On grade (1 if yes, 0 otherwise) [2016–2021]	–0.485	–2.62	0.0409	–0.0409
Four-leg intersection (1 if yes, 0 otherwise) [2016]	–0.623	–2.63	0.0586	–0.0586
Four-leg intersection (1 if yes, 0 otherwise) [2017–2018]	0.395	2.36	–0.0434	0.0434
Four-leg intersection (1 if yes, 0 otherwise) [2019–2023]	–0.291	–2.61	0.0218	–0.0218
U-turn with deceleration lane only (1 if yes, 0 otherwise) [2018]	1.203	2.84	–0.1536	0.1536
Environmental characteristics				
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2020]	1.110	11.86	–0.1256	0.1256
Dark, no street lighting (1 if yes, 0 otherwise) [2021–2023]	0.864	6.67	–0.0733	0.0733
Temporal characteristics				
New Year festival period (1 if yes, 0 otherwise) [2016]	–0.493	–2.57	0.0478	–0.0478
New Year festival period (1 if yes, 0 otherwise) [2019–2023]	–0.410	–3.78	0.0301	–0.0301
Region				
Eastern region (1 if yes, 0 otherwise) [2016]	0.786	4.45	–0.0909	0.0909
Eastern region (1 if yes, 0 otherwise) [2017–2021]	0.433	5.00	–0.0412	0.0412
Eastern region (1 if yes, 0 otherwise) [2022–2023]	0.727	6.26	–0.0571	0.0571
Northeastern region (1 if yes, 0 otherwise) [2016]	0.879	6.28	–0.1013	0.1013
Northeastern region (1 if yes, 0 otherwise) [2017–2018]	0.479	4.82	–0.0524	0.0524
Northeastern region (1 if yes, 0 otherwise) [2019–2023]	0.712	9.84	–0.0615	0.0615
Northern region (1 if yes, 0 otherwise) [2020–2023]	–0.408	–3.47	0.0290	–0.0290
Southern region (1 if yes, 0 otherwise) [2016–2020]	0.485	5.73	–0.0492	0.0492
Western region (1 if yes, 0 otherwise) [2016]	0.718	3.50	–0.0827	0.0827
Western region (1 if yes, 0 otherwise) [2017–2018]	0.280	1.75	–0.0302	0.0302
Western region (1 if yes, 0 otherwise) [2019–2023]	0.705	6.60	–0.0618	0.0618
Heterogeneity in the means of random parameters				
One truck with more than 10 wheels involved [2016–2023]: Evening [2016–2023]	0.568	4.37		
Horizontal curve [2017–2023]: Raining [2020–2023]	–1.328	–3.82		
Two or more pickup trucks involved [2016–2023]: Peak hours [2016–2023]	0.396	3.13		
Heterogeneity in the variances of random parameters				
One truck with more than 10 wheels involved [2016–2023]: U-turn with deceleration lane only [2019–2023]	1.093	2.47		
Model statistics				
Number of observations			33,646	

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Number of estimated parameters		81		
Log-likelihood at zero, LL(0)		−23321.63		
Log-likelihood with constant only, LL(C)		−15177.81		
Log-likelihood at convergence, LL(β)		−12598.28		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.460		
AIC / BIC		25,358.6 / 26,040.9		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure from the partially constrained procedure applied inside the random parameters model: adjacent periods merged when a likelihood ratio test (simulated log-likelihood, 2,000 Halton draws) does not reject equal parameters at the 90% level. The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = −(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.